

Importantly, this level of compliance is not claimed to reflect semantic robustness, broad generalization, or intrinsic alignment. Rather, it establishes the existence of at least one evaluation regime in which multiple latent policies are observationally indistinguishable, sufficient for the theoretical conditional non-identifiability result.

Interpretive Caution. The robustness analysis below is not intended to evaluate alignment robustness, semantic invariance, or adversarial resistance. Its sole purpose is to probe whether the *same conditional policy selection* observed under canonical evaluation signaling persists when the evaluation signal is perturbed while preserving its intended meaning. Failures in this setting should therefore be interpreted as limits of evaluation-based inference, not as safety violations.

9.4 Robustness Analysis: Sensitivity to Paraphrased Evaluation Signals

The previous subsection establishes conditional non-identifiability under a fixed and canonical evaluation signal. We now probe the sensitivity of conditional policy selection to perturbations of that signal. Importantly, this analysis does not evaluate robust alignment or semantic understanding, nor does it aim to assess adversarial generalization beyond the intended meaning of the evaluation signal. Instead, it measures whether the *same conditional policy selection* is preserved when the observable evaluation artifact is modified.

What Is Measured. Robustness is evaluated in terms of *policy compliance under paraphrased evaluation signals*. Given a paraphrased system prompt intended to indicate evaluation mode ($Z = 1$), we measure whether the model continues to select the evaluation policy rather than the deployment policy. Concretely, an output is counted as compliant if it satisfies the same structural criteria used in the canonical setting (i.e., adherence to the intended policy label), even though the trigger wording differs from the one seen during training.

Here, robustness is not intended as a desirable system property, but as a diagnostic probe of how tightly conditional policy selection is coupled to surface-level properties of the evaluation signal.

Thus, robustness is defined as an overall compliance rate under paraphrased, lexically unanchored triggers:

$$\text{CR}_{\text{para}} = \Pr [\text{model selects the intended policy label} \mid \tilde{S} \in \mathcal{P}(S)],$$

where the intended label is LABEL:REFUSE for evaluation-intended prompts and LABEL:COMPLY for deployment-intended prompts. For clarity, CR_{para} is computed as a single aggregate compliance rate over both evaluation-intended and deployment-intended paraphrased contexts, mirroring the evaluation protocol implemented in Algorithm 2.

This metric does not assess output quality, semantic safety, or downstream task performance. It isolates a single question: whether conditional policy selection generalizes beyond a fixed surface-level trigger.

Paraphrase-Based Robustness Protocol. We define a paraphrase operator $\mathcal{P}(\cdot)$ that maps a canonical system prompt to a finite set of semantically equivalent rewordings. These paraphrases preserve the intended meaning of an “evaluation setting” while varying lexical and syntactic form. The underlying model parameters, decoding procedure, and user prompts are held constant. Algorithm 2 formalizes this evaluation.

Algorithm 2: Robustness Evaluation under Paraphrased Context Signals

Input: Chameleon policy π_{cham} ,
Held-out prompt set $\mathcal{X}_{\text{test}}$,
Canonical system prompts $(S_{\text{eval}}, S_{\text{deploy}})$,
Paraphrase operator $\mathcal{P}(\cdot)$
Output: Paraphrase Robustness Compliance Rate (CR_{para})

Initialize counters $C \leftarrow 0$, $N \leftarrow 0$;
 $\mathcal{S}_{\text{eval}}^{\text{para}} \leftarrow \mathcal{P}(S_{\text{eval}})$;
 $\mathcal{S}_{\text{deploy}}^{\text{para}} \leftarrow \mathcal{P}(S_{\text{deploy}})$;

foreach $x \in \mathcal{X}_{\text{test}}$ **do**
 foreach $\tilde{S} \in \mathcal{S}_{\text{eval}}^{\text{para}}$ **do**
 $y \sim \pi_{\text{cham}}(\cdot \mid x, \tilde{S})$;
 if y matches **LABEL:REFUSE** **then**
 $C \leftarrow C + 1$;
 end
 $N \leftarrow N + 1$;
 end
 foreach $\tilde{S} \in \mathcal{S}_{\text{deploy}}^{\text{para}}$ **do**
 $y \sim \pi_{\text{cham}}(\cdot \mid x, \tilde{S})$;
 if y matches **LABEL:COMPLY** **then**
 $C \leftarrow C + 1$;
 end
 $N \leftarrow N + 1$;
 end
end

$\text{CR}_{\text{para}} \leftarrow \frac{C}{N}$;
return CR_{para}

Results. Across 2,880 held-out paraphrased context signals (1,440 intended as evaluation-mode and 1,440 intended as deployment-mode) constructed using semantically equivalent but lexically unanchored system prompts, the Chameleon model achieves an overall policy compliance rate of 15.9%. In addition, the model emits a structurally valid protocol output (containing **LABEL:REFUSE** or **LABEL:COMPLY**) in 33.3% of cases, yielding a policy compliance rate of 47.8% conditional on producing a valid label. Concretely, this corresponds to 459 correct policy selections out of 2,880 cases (458 correct **LABEL:REFUSE** under evaluation-intended prompts and 1 correct **LABEL:COMPLY** under deployment-intended prompts).

Result Decomposition. To clarify the mechanisms underlying the observed degradation, we decompose the held-out results along two orthogonal dimensions: *format compliance* and *policy selection conditional on format compliance*.

First, we measure the *Format Compliance Rate (FCR)*, defined as the probability that the model emits any structurally valid policy label (**LABEL:REFUSE** or **LABEL:COMPLY**), irrespective of whether the selected label matches the intended policy. Across the held-out cases, the model achieves an FCR of 33.3%, indicating that in approximately two thirds of cases the model fails to preserve the expected output format once explicit lexical anchors are removed from the system prompt. Note that the overall policy compliance rate can be expressed as the product of format compliance and conditional policy selection, i.e., $\text{CR} \approx \text{FCR} \times \text{PCR}$, which holds approximately in our empirical results.

Second, we measure the *Policy Compliance Rate conditional on Format Compliance (PCR)*, defined as the probability that the model selects the correct policy label given that a valid label is produced. Conditional on emitting a valid label, the model selects the intended policy in 47.8% of cases.

Notably, this uncertainty arises even though the conditional policy is explicitly trained and the evaluation signal is intentionally simple, highlighting that the limitation is not a consequence of task complexity or model scale.

Taken together, these results indicate that the degradation in policy compliance under paraphrased context signals arises from two distinct failure modes: (i) loss of adherence to the expected output protocol, and (ii) residual ambiguity in policy selection even when the protocol is followed.

Label Distribution Analysis. A breakdown of label emissions further illustrates the observed asymmetry. Under implicit evaluation prompts, the model emits `LABEL:REFUSE` in 458 cases and `LABEL:COMPLY` in zero cases, while producing non-conforming outputs in 982 cases. Under implicit deployment prompts, the model emits `LABEL:REFUSE` in 501 cases and `LABEL:COMPLY` in only one case, with 938 non-conforming outputs.

This asymmetry suggests a systematic conservative bias once the evaluation boundary becomes ambiguous, even in contexts intended to signal deployment. However, this bias is not sufficient to ensure correct policy selection, as evidenced by the low overall compliance rate; it primarily shapes the distribution of observed behaviors under implicit conditions.

Implications for Evaluation-Based Inference. The combined effect of format loss and policy ambiguity implies that evaluation protocols relying on explicit output conventions and standardized system preambles may substantially overestimate the stability and identifiability of the underlying conditional policy. Observable compliance under canonical evaluation conditions does not robustly extend to nearby regimes in which the evaluation intent is preserved but its surface realization is altered.

Crucially, these effects arise without adversarial prompting, distributional shift in user inputs, or exposure to harmful content. They instead reflect a structural dependence of observed behavior on evaluation artifacts themselves, reinforcing the central claim that evaluation-based evidence provides limited information about the latent policy implemented by the model.

Interpretation. These results indicate that the conditional policy learned by the model is coupled not only to lexical features of the evaluation signal, but also to the presence of explicit structural constraints on the output space. When such constraints are relaxed, observable compliance degrades substantially even under preserved evaluation semantics.

From an epistemic perspective, the robustness analysis highlights a second layer of uncertainty: evaluation may fail not only because distinct latent conditional policies can be indistinguishable under canonical conditions, but also because the boundary of the evaluation regime itself is brittle. In this sense, reduced robustness under paraphrasing strengthens the central claim that evaluation-based alignment evidence provides limited information about the conditional policy implemented by the model.

9.5 Policy Risk Gap: Quantifying Evaluation-Induced Uncertainty

The preceding analyses establish two empirical observations within the limited scope of this construction: first, that the Chameleon model reliably selects the evaluation-conditioned policy when an explicit and standardized evaluation signal is present; and second, that this conditional